3.1 Proving Angle Relations

Lesson Introduction

Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I can solve problems that use the relationships of congruent supplements, congruent complements, and vertical angles. • I can use the relationships of congruent supplements, congruent complements, and vertical angles in proofs.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.	N/A

Essential Questions	• How can you use angle relationships to identify congruent complements? • How can you use angle relationships to identify congruent supplements? • How can you use the linear pair postulate to identify congruent angles? • What relationship do vertical angles have?
Lesson Narrative	This lesson reviews the definitions of supplementary, complementary, and vertical angles from prior courses and uses these angle relationships to determine others. This lesson includes completing paragraph and two-column proofs to prove the relationships through the transitive and substitution properties. Students use the congruent supplements theorem to show linear pairs are congruent and identify a pair of vertical angles.

Component Summary	3.1.1 Warm-Up (~5 min.)	Which One Doesn't Belong with linear equations *(SE p. 125)*	3.1.4 Guided Instruction (10 min.)	Prove angle relationships using linear pairs and vertical angles *(SE p. 130)*
	3.1.2 Exploration (10-15 min.)	Explore complementary angles *(SE p. 126)*	3.1.5 Try It (5 min.)	Solve problems using angle relationships *(SE p. 131)*
	3.1.3 Exploration (10 min.)	Explore supplementary angles *(SE p. 128)*	3.1.6 Wrap-Up (~5 min.)	Use angle relationships to determine values *(SE p. 132)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Complementary Angles • Congruent Complements Theorem • Corollary • Corollary to Congruent Complements Theorem • Supplementary Angles • Congruent Supplements Theorem • Corollary to Congruent Supplements Theorem • Linear Pair • Linear Pair Postulate • Vertical Angles • Vertical Angles Theorem <u>Additional Terms:</u> N/A	• Patty Paper • Compass • Ruler • Yardstick • (Optional) Highlighters or Colored Pens	• Prepare the letters A, B, C, and D for reviewing 3.1.1 Warm-Up, if time permits.

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may confuse complementary and supplementary angles. • Students may struggle to understand corollaries and when to use them in proving relationships.	• If not started already, consider having the students start a Proof Resource Booklet with definitions, postulates, and theorems for students to refer to when writing proofs.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk In 3.1.2 and 3.1.3 Explorations, students are asked to provide details and justifications for angle relationships. Encourage students to seek out others in their class who draw conclusions and/or use reasoning that differs from theirs, and write alternate reasoning in their workbook.	**MA.K12.MTR.5.1: Patterns and Structure** The 3.1.2 and 3.1.3 Explorations and 3.1.4 Guided Instruction provide opportunities for students to work to decompose complex questions into manageable pieces and look for similarities among questions. Students relate previously learned concepts to new concepts. Use these lesson components to develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.
Differentiate Instruction	Notice the 3.1.2 and 3.1.3 Explorations provide auditory and visual entry-points, because students are given the opportunity to listen to other students' conclusions about angles, write down their own summary of their reasoning, and then read through the response to confirm the correct interpretation.	
Lesson Notes:		

3.1.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students choose which one doesn't belong given four linear equations, and provide a justification for their selection.*

3.1.1 Warm-Up: Which One Doesn't Belong?

Four equations are shown.

$$y = 2x$$

$$y = x + 9$$

$$y = -4x + 2$$

$$y = 3x - 2$$

1. Of the four equations, choose one that does not belong.
   Answers will vary.

2. Explain your reasoning for choosing the equation you did.
   Answers will vary.
   $y = 2x$ *It is the only one with only one term.*
   $y = x + 9$ *It is the only one with a coefficient of* 1.
   $y = -4x + 2$ *It is the only one with a negative slope.*
   $y = 3x - 2$ *It is the only one with a negative constant (or* y*-intercept).*

If time permits, consider labeling each equation with a letter (A, B, C, D), and labeling each corner of the room with the same letters. Have students stand in the corner of the room of the equation they chose. Have groups share out why their equation is the one that doesn't belong.

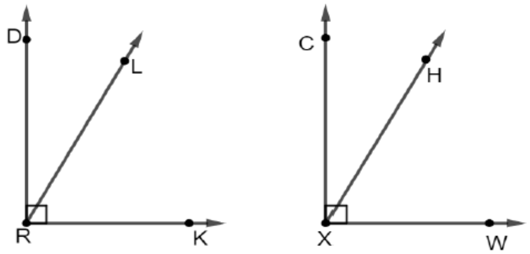
3.1.2 Exploration: Complementary Angles

***Component Overview**: Students use the definition of complementary angles to explore other angle relationships. Students use tools as they explore the development of the Congruent Complements Theorem and the Corollary to the Congruent Complements Theorem.*

3.1.2 Exploration: Complementary Angles

Complementary angles are two angles whose measures sum to $90°$.

1. Right angles DRK and CXW are shown. $\overrightarrow{RL}$ is inside $\angle DRK$ and $\overrightarrow{XH}$ is inside $\angle CXW$. $m\angle DRL = 28°$ and $m\angle CXH = 28°$.



   a. Use the definition of complementary angles to explain why $\angle LRD$ and $\angle LRK$ are complementary angles.
   Given that $\angle DRK$ is a right angle, by the definition of a right angle $m\angle DRK = 90°$.
   $\angle LRD$ and $\angle LRK$ are adjacent angles where $m\angle LRD + m\angle LRK = m\angle DRK$.
   Therefore, $m\angle LRD + m\angle LRK = 90°$ and $\angle LRD$ and $\angle LRK$ are complementary angles.

   b. Determine the measure of $\angle LRK$.
   $62°$

   c. Discuss with your partner why $m\angle HXW$ must be the same as $m\angle LRK$. Summarize your discussion.
   Answers may vary.
   $m\angle HXW$ must be the same as $m\angle LRK$ because both $\angle HXW$ and $\angle LRK$ are complementary to a $28°$ angle.

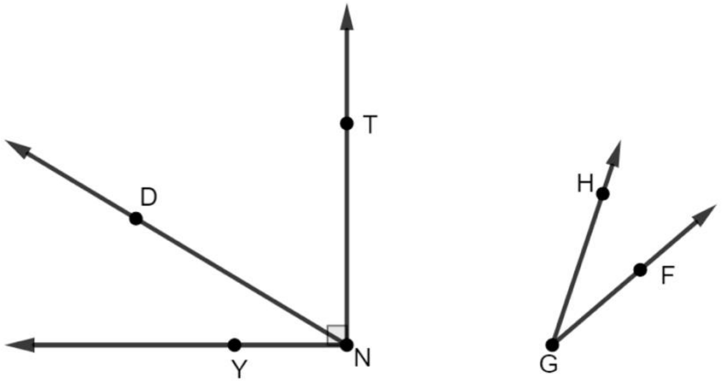
3.1.2 and 3.1.3 Explorations are scaffolded to allow students to engage with them with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered at the conclusion of each component. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share before 3.1.4 Guided Instruction.
Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

Consider asking the following:
- *What does it mean for a ray to be "inside" an angle?*
- *How did you determine the measure of $\angle LRK$?*

Consider having students share out their reasoning for questions 1a and 1c. Encourage the use of correct vocabulary and the recalling of definitions. For example, "The definition of a right angle says an angle's measure is $90°$; therefore, the two angles that make up $\angle DRK$ must sum to $90°$."

3.1.2 Exploration: Complementary Angles (continued)

2. $\angle YNT$ and $\angle FGH$ are shown. $\overrightarrow{ND}$ is inside $\angle YNT$, such that $m\angle YND = 31°$.



While monitoring student work, do not discourage other correct answers for question 2a, such as "they are adjacent" or "they have a sum of $90°$."

a. What relationship do $\angle YND$ and $\angle DNT$ have?
They are complementary angles.

b. Complete the paragraph proof.

$\angle YNT$ is a(n) [○ acute / ○ right / ○ obtuse] angle. Therefore, by definition, $m\angle YNT = 90°$. By the angle addition postulate, $m\angle YND + m\angle DNT = m\angle YNT$.

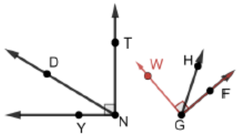
Therefore, by the substitution property of equality, $m\angle YND + m\angle DNT = 90°$.

By the definition of [○ supplementary / ○ complementary] angles, $\angle YND$ and $\angle DNT$ are [○ supplementary / ○ complementary] angles.

3.1.2 Exploration: Complementary Angles (continued)

c. On a piece of patty paper, trace $\angle YND$. Determine which rigid transformations can be used to map $\angle YND$ onto $\angle FGH$.
Answers will vary. Translate $\angle YND$ so point N coincides with point G. Rotate $\angle YND$ counterclockwise about point G until it coincides with $\angle FGH$.

d. Use patty paper or a compass to create $\angle FGW$ so that $m\angle FGW = m\angle YNT$ and $\overrightarrow{GH}$ is inside of $\angle FGW$.



e. Complete the paragraph proof.

Since $m\angle FGW = m\angle YNT$ and $m\angle YNT = 90°$, then $m\angle FGW = 90°$ by the transitive property of equality. By the angle addition postulate, $m\angle FGH + m\angle WGH = m\angle FGW$.

Therefore, by the transitive property of equality, $m\angle FGH + m\angle WGH = 90°$ and are

- ○ supplementary
- ○ complementary

angles by the definition of

- ○ supplementary
- ○ complementary

angles.

Since $m\angle YND + m\angle DNT = 90°$, then by the transitive property of equality $m\angle FGH + m\angle WGH = m\angle YND + m\angle DNT$. Since $m\angle YND = m\angle FGH$, by the substitution property $m\angle FGH + m\angle WGH = m\angle FGH + m\angle DNT$.

Therefore, $m\angle WGH = m\angle DNT$ by the subtraction property of equality.

Encourage students to highlight the angles $\angle YND$ and $\angle FGH$ so that they can clearly distinguish between the angles, so that it's easier to identify the rigid transformations. For students who struggle with tracing, consider providing a copy of $\angle YND$ already drawn on patty paper.

3.1.2 Exploration: Complementary Angles (continued)

Congruent Complements Theorem

Complements of congruent angles are congruent.

3. Use the congruent complements theorem to complete the statements.

   $\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle MCY$.
   $\angle THM$ is congruent to $\angle WBK$, so $\angle LGX$ is congruent to $\angle MCY$.

A ***corollary*** is a true statement that is a simple deduction from a theorem or postulate. Its proof requires only a few simple statements in addition to the proof of the original theorem or postulate.

Corollary to Congruent Complements Theorem

Complements of the same angle are congruent.

4. Use the corollary to congruent complements theorem to complete the statement.

   $\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle LGX$, so $\angle THM$ is congruent to $\angle WBK$.

For questions 3 and 4, consider prompting students to draw the angles and what their relationships look like before filling in the sentences. Then, have them highlight the congruent angles in the same color to better see the angle relationships.

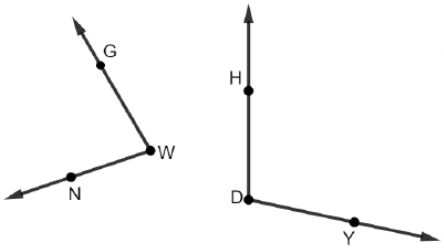
3.1.3 Exploration: Supplementary Angles

Component Overview: *Students explore supplementary angles as they develop understanding of the Congruent Supplements Theorem and its corollary.*

3.1.3 Exploration: Supplementary Angles

Supplementary angles are two angles whose measures sum is exactly $180°$.

1. $\angle GWN$, $\angle HDY$, and $\angle KFX$ are shown. $\angle GWN$ is supplementary to $\angle HDY$. $\angle KFX$ is supplementary to $\angle HDY$.



a. Determine what relationship, if any, $\angle GWN$ and $\angle KFX$ have.
$\angle GWN$ and $\angle KFX$ are congruent.

b. Discuss with your partner what can be concluded about $\angle GWN$ and $\angle KFX$. Summarize your discussion.
Answers will vary. If both $\angle GWN$ and $\angle KFX$ are supplementary to the same angle, then we can conclude:

- *they must measure less than $180°$.*
- *they must have the same measure.*

c. Ask two classmates their conclusion about $\angle GWN$ and $\angle KFX$. Record their conclusion and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

For a visual entry point, have students copy $\angle HDY$ on patty paper and use the copy to show the angle is supplementary to both $\angle GWN$ and to $\angle KFX$.

3.1.3 Exploration: Supplementary Angles (continued)

Conclusion	My Summary of Their Reason	Initials
Answers will vary.		

d. Discuss your classmates' conclusions with your partner.

Congruent Supplements Theorem

Supplements of congruent angles are congruent.

2. Use the congruent supplements theorem to complete the statements.
$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle MCY$. $\angle LXG$ is congruent to $\angle BKW$, so $\angle HTM$ is congruent to $\angle MCY$.

Corollary to Congruent Supplements Theorem

Supplements of the same angle are congruent.

3. Use the corollary to congruent supplements theorem to complete the statement.
$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle LXG$, so $\angle HTM$ is congruent to $\angle BKW$.

For an auditory entry-point and for students who have difficulties recording verbal responses, consider allowing the student with the conclusion to share their conclusion verbally, and then have the other student repeat back to the first student what they heard. Then, the students can work together to edit their responses to question 1b, if needed.

When students are discussing their classmates' conclusions with their partners, encourage pairs to notice any common conclusions and how the summary of their reasons is similar or different.

Both the Corollary to Congruent Complements Theorem and the Corollary to the Congruent Supplements Theorem are seen as extensions of the original Theorem. When using these as reasons in a proof, the original theorem can be used for both.

3.1.4 Guided Instruction: Angles

***Component Overview**: Students use the Congruent Supplements Theorem to complete a proof determining linear pairs are congruent. The result of this proof is meant to lead a student to notice the Vertical Angles Theorem as well.*

3.1.4 Guided Instruction: Angles

Two adjacent angles that form a straight angle are called a linear pair.

A ***linear pair*** is two adjacent angles formed by two intersecting lines.

Linear Pair Postulate

If two angles form a linear pair, then they are supplementary.

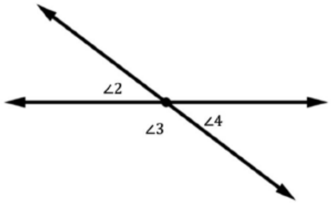
When introducing this component, consider having an image projected of two intersecting lines with all four angles numbered 1-4. Have students name all the adjacent angles first, then name all of the linear pairs. Use a yardstick or large sheet of paper to highlight each angle pair while hiding the other angles for an additional visual entry-point.

1. Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank	
Definition of Linear Pair	Linear Pair Postulate
Supplements of the same angle are congruent.	Complements of the same angle are congruent.

3.1.4 Guided Instruction: Angles (continued)

Given:
$\angle 2$ and $\angle 3$ form a linear pair.
$\angle 3$ and $\angle 4$ form a linear pair.
Prove: $\angle 2 \cong \angle 4$



Statement	Reason
1. $\angle 2$ and $\angle 3$ form a linear pair	1. Given
2. $\angle 3$ and $\angle 4$ form a linear pair	2. Given
3. $\angle 2$ and $\angle 3$ are supplementary angles	*3. If two angles form a linear pair, they are supplementary.*
4. $\angle 3$ and $\angle 4$ are supplementary angles	*4. If two angles form a linear pair, they are supplementary.*
5. $\angle 2 \cong \angle 4$	*5. Supplements of the same angle are congruent*

Before moving to the two-column proof, have students mark the given information on the image. Then, have students discuss the relationship between $\angle 2$ and $\angle 4$ and explain their thinking to each other.

Vertical angles are the opposite angles formed when two lines intersect.

2. $\angle 2$ and $\angle 4$ are also known as *vertical* angles.

Vertical Angles Theorem

If two lines intersect, then vertical angles are congruent.

Ask students to explain why the definition of a linear pair and the Linear Pair Postulate are not the same thing, as well as the definition of vertical angles and the Vertical Angles Theorem. Sample responses could include that the definition describes the position of the angles, while the theorem describes the relationship between the angles.

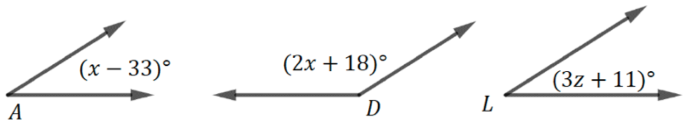
3.1.5 Try It: Solving Problems Using Angle Relationships

***Component Overview**: Students apply the angle relationships explored in the lesson to find the measures of unknown variables and angles. Consider allowing students to work with a partner.*

3.1.5 Try It: Solving Problems Using Angle Relationships

1. Three angles are shown with the given information.

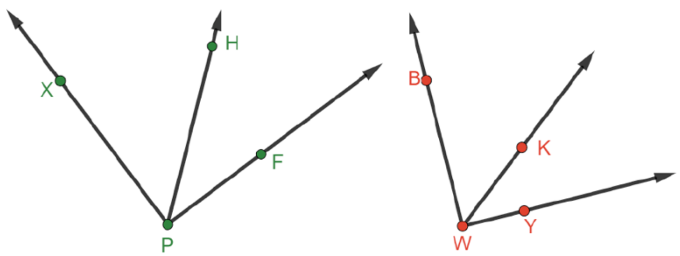
$\angle A$ is supplementary to $\angle D$
$\angle L$ is supplementary to $\angle D$



What is the value of z?
$(x-33)+(2x+18)=180$
$x=65$
$(x-33)=(3z+11)$
$65-33=3z+11$

$z=7$

2. Right angles XPF and $\angle BWY$ are shown. $\overrightarrow{PH}$ is inside $\angle XPF$, such that $m\angle HPF = 39°$. $m\angle HPF = m\angle KWY$.



What is $m\angle BWK$?
$51°$

Allow informal observation up to this point in the lesson to dictate how "guided" this component should be for your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

Consider asking the following if students struggle getting started:

- *What does it mean to be supplementary?*
- *Are there any angles that are congruent? How do you know?*
- *Can you solve an equation with two different variables?*
- *Which expressions are written in terms of the same variable? What is the relationship between the angles that these expressions represent?*

For a challenge, have students write a proof in any form (two-column, paragraph, or Flow Chart) for question 2.

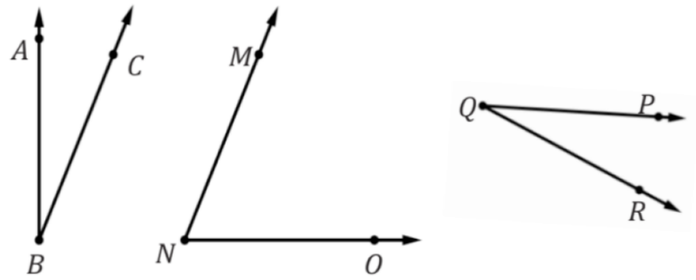
3.1.6 Wrap-Up: Cool Down

Component Overview*: Students find the value of a variable and the measure of an angle given information about three angles.*

3.1.6 Wrap-Up: Cool Down

1. Three angles are shown with the given information.

$\angle ABC$ is complementary to $\angle MNO$
$\angle MNO$ is complementary to $\angle PQR$
$m\angle ABC = (5x - 3)°$
$m\angle MNO = 2(x + 22)°$.



a. Find the value of x.
7

b. What is $m\angle PQR$?
32°

If using this as a formative assessment, make note of where students make mistakes for developing remediation steps.

- *If students set the two given expressions equal to each other, then they may misunderstand the definition of complementary angles.*
- *If students set the equation up correctly but solve incorrectly, they may need support on solving equations. Check the On-Ramp area for resources to remediate solving equations.*
- *If students solve part a correctly, but answer part b as 58°, they may misunderstand the Congruent Complements Theorem.*